HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines for ETL processes, handling approximately **500,000 records** daily for downstream analytics and reporting.
- Developed and maintained **Java Spring Boot** microservices, including **REST APIs** for inter-service communication and feature enhancements, contributing to a **15%** reduction in API latency.
- Managed production support for enterprise applications, addressing P1/P2 incidents and ensuring adherence to strict **SLA** targets using **Splunk** and **Dynatrace** for monitoring and root cause analysis.
- Applied machine learning techniques using **scikit-learn** and **pandas** to build and evaluate models for ad relevance scoring, achieving a lift of **8%** in click-through rates on held-out test data.
- Collaborated with product managers and data scientists to prototype, optimize, and productionize **large-scale ML models** for the ads funnel, focusing on retrieval and ranking algorithms.
- Wrote and maintained unit tests using **JUnit** for **Java** components, ensuring code quality and achieving over **85%** test coverage for new feature development.
- Containerized ML services using **Docker** and deployed them to **AWS** environments, configuring basic resource limits and networking for predictable performance.

SKILLS

Programming Languages – Python, Java, SQL, R, bash

ML/Data Science – scikit-learn, pandas, NumPy, ML model training, inference, A/B testing, statistical analysis, data modelling, DAX, Power Query

Frameworks & Libraries – Spring Boot, Flask, FastAPI, Hibernate, JPA, React, Angular

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, Terraform

Databases – PostgreSQL, MySQL, Oracle

Tools & Platforms – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau, RStudio

PROJECTS

ML Model for Search Ad Relevance

- Developed a **Python**-based feature engineering pipeline to extract user and listing signals from raw transaction data for an ad relevance model, processing approximately **1.5 TB** of historical data.
- Trained and evaluated a **gradient-boosted classifier** (LightGBM) on a dataset of **~750,000 ad impressions**, optimizing for click-through rate prediction and achieving an AUC of **0.82** on a held-out validation set.
- Packaged the trained model behind a **FastAPI** endpoint, implementing input validation and logging for feature drift detection, supporting an average throughput of **150 requests/second**.

Etsy-Style Marketplace Search Ranking System

- Designed and implemented a **Java Spring Boot** service that integrates with a search index, responsible for retrieving and ranking potential ad placements based on user query and listing attributes.
- Built **REST APIs** to serve ranking scores to the ad allocation system, handling approximately **2,000 requests per minute** during peak hours.
- Collaborated with an analytics partner to define and implement metrics for A/B testing new ranking algorithms, observing a measurable impact on buyer engagement.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)